

## **TABLE OF CONTENTS**

### **E-421 CURRENT TRANSFORMERS**

|                                                        | Page No. |
|--------------------------------------------------------|----------|
| 1. SCOPE OF WORK.....                                  | E-421-2  |
| 2. CODES AND STANDARD .....                            | E-421-2  |
| 2.1 Indonesian National Standards.....                 | E-421-2  |
| 2.2 International Electrotechnical Commission .....    | E-421-2  |
| 2.3 National Electrical Manufacturers Association..... | E-421-2  |
| 3. OPERATING AND DESIGN CONDITIONS .....               | E-421-2  |
| 4. DETAILED REQUIREMENTS .....                         | E-421-2  |
| 4.1 General.....                                       | E-421-2  |
| 4.2 Ratings .....                                      | E-421-3  |
| 4.3 Oil-Filled Current Transformers .....              | E-421-4  |
| 4.4 Corona.....                                        | E-421-5  |
| 4.5 Porcelain .....                                    | E-421-5  |
| 4.6 Nameplates .....                                   | E-421-5  |
| 5. PERFORMANCE TESTING AND GUARANTEES.....             | E-421-5  |
| 5.1 Type Tests.....                                    | E-421-5  |
| 5.2 Routine Tests .....                                | E-421-6  |
| 6. QUALITY ASSURANCE/QUALITY CONTROL .....             | E-421-6  |

## **E-421 CURRENT TRANSFORMERS**

### **1. SCOPE OF WORK**

The work shall include the design, supply, manufacturing, factory testing, delivery to site, installation, commissioning and testing of current transformers included in this Contract in accordance with Part 4, Subsection 1.2 Summary of Work.

### **2. CODES AND STANDARD**

The Contractor shall comply with the requirements of Part 4, Subsection 2.5 Codes and Standards, and the following specific standards:

#### **2.1 Indonesian National Standards**

PUIL : Peraturan Umum Instalasi Listrik – Indonesia.

SPLN : Standard Perusahaan Umum Listrik Negara.

#### **2.2 International Electrotechnical Commission**

IEC 60044 : Instrument transformers

IEC 60060 : High voltage test techniques

IEC 60270 : High-voltage test techniques – Partial discharge measurements

IEC 62155 : Hollow pressurized and unpressurized ceramic and glass insulators for use in electrical equipment with rated voltages greater than 1000V

#### **2.3 National Electrical Manufacturers Association**

NEMA 107 : Method of Measurements of Radio Influence Voltage (RIV) of High Voltage Apparatus

### **3. OPERATING AND DESIGN CONDITIONS**

The substation current transformers shall be designed for outdoor operation on an effectively grounded system located in a tropical climate along the seacoast.

### **4. DETAILED REQUIREMENTS**

#### **4.1 General**

The specified accuracy and burden ratings are the minimum requirements. The Contractor shall verify the ratings based on the application and his protective re-

lays and instruments and propose the appropriate ratings.

Current transformers secondary circuits shall be grounded at only one (1) point, preferable at the first panel entry at the control building.

Current transformer assemblies shall be provided with a terminal box for secondary winding connections suitable furnished with shorting type terminal blocks for outgoing cable connections.

The current transformer secondary transient characteristic shall be the best suitable for the protection schemes provided by the Contractor. Type TPX current transformer are preferred, however, Contractor may offer either type TPY or TPZ type current transformers to match his relaying system performance characteristics.

#### 4.2 Ratings

- |     |                                                                                                                      |                                                                                                       |           |
|-----|----------------------------------------------------------------------------------------------------------------------|-------------------------------------------------------------------------------------------------------|-----------|
| (1) | Nominal system voltage .....                                                                                         | 500 kV                                                                                                | Neutral   |
| (2) | Rated maximum system voltage .....                                                                                   | 550 kV                                                                                                | 36 kV     |
| (3) | Frequency .....                                                                                                      | 50 Hz                                                                                                 | 50 Hz     |
| (4) | Short-time current:                                                                                                  |                                                                                                       |           |
|     | a. Short-time thermal current (I <sub>th</sub> ).....                                                                | 40 kA                                                                                                 | 40 kA     |
|     | b. Short-time dynamic current (I <sub>dyn</sub> ) .....                                                              | 100 kA                                                                                                | 100 kA    |
| (5) | Ratio .....                                                                                                          | *)                                                                                                    | *)        |
|     | Note : *) The Contractor shall confirm for metering and relaying system proposed and shall be approved by the Owner. |                                                                                                       |           |
| (6) | Secondary ratings:                                                                                                   |                                                                                                       |           |
|     | a. Burden .....                                                                                                      | As required for the application including margin of 20% for spare capacity (min. 30 VA at each ratio) |           |
|     | b. Accuracy                                                                                                          |                                                                                                       |           |
|     | c.1 Protective Relaying.....                                                                                         | TPX                                                                                                   | TPX       |
|     | c.2 Instrument and metering.....                                                                                     | Class 0.5                                                                                             | Class 0.5 |
| (7) | Phase-to-ground insulation                                                                                           |                                                                                                       |           |
|     | a. Impulse withstand voltage (1.2 x 50 μs) crest .....                                                               | 1,550 kV                                                                                              | 170 kV    |

- b. 50 Hz dry (and wet\*) withstand test voltage..... 680(-) kV 70(70) kV
- c. Switching impulse withstand test voltage ..... 1,175 kV N/A
- (8) Radio Interference Voltage (RIV)  
The maximum RIV at 1 MHz shall be:
  - a. Applied phase to ground voltage ..... 334 kV N/A
  - b. R.I.V. microvolts ..... 2,500  $\mu$  volts N/AThe measurements shall be in accordance with NEMA 107 procedure. In making the RIV test, radio noise readings shall be taken at various points so as to determine the break points.
- (9) Pollution level ..... Very heavy
- (10) Partial Discharge Test:
  - a. Applied phase to ground voltage ..... 550 kV N/A
  - b. Maximum discharge ..... 10 picocoulombs N/A

### **4.3 Oil-Filled Current Transformers**

#### **4.3.1 General**

Current transformer may be of the bar, single or multi turn primary and shall be hermetically sealed. Oil used to fill current transformer (if any) shall meet the requirements specified in Technical Specification E-001.

#### **4.3.2 Accessories**

Current transformers shall have the following accessories for monitoring the oil:

- (1) Oil level gauge visible from ground level.
- (2) Oil drain valve with sampling means.
- (3) Current transformers shall have automatic pressure relief device with an alarm device for remote annunciator input.

#### **4.3.3 Grounding Pads**

Two (2) grounding terminal pads on diagonal opposite locations shall be furnished for connection to the ground grid with an ampacity not less than that of 240 mm<sup>2</sup> copper cable. Standard pads shall be furnished for the connection to the ground grid

#### **4.4 Corona**

Surfaces in contact with line potential shall contain no projection points or irregularities and they shall be designed for a minimum of visible corona or brush discharge when the transformer is completely assembled.

No visual corona shall exist at applied voltage. Lights shall be turned off in the test bay and visual start and stop corona voltages shall be recorded with the corresponding radio noise voltages on both ascending and descending voltage values. Photographs shall be taken for each recorded voltage value. NEMA 107 test procedures shall be used.

#### **4.5 Porcelain**

The Contractor shall meet the requirements of IEC 62155 for all porcelain used in these transformers.

Porcelain insulation for current transformers shall be chocolate brown color and shall be permanently marked.

#### **4.6 Nameplates**

Nameplates as per IEC 60044 shall be provided clearly identifying individual winding classification for multiple transformer units.

### **5. PERFORMANCE TESTING AND GUARANTEES**

#### **5.1 Type Tests**

If type tests for each transformer of identical design and the test performed in the following sequence has been done, then no repeated test shall be done, however type test certificate shall be submitted for approval.

- (1) Short time current test shall be applied as per IEC 60044.
- (2) Temperature rise test shall be applied as per IEC 60044.
- (3) Lightning impulse voltage withstand test shall be applied as per IEC 60044.
- (4) Power frequency withstand test as per IEC 60060
- (5) Radio influence voltage test and corona test shall be on a transformer as assembled in service in accordance with NEMA 107 along with corona tests as per this Specification.
- (6) Partial discharge test shall be applied preferably in accordance with IEC

60270.

- (7) Dielectric loss angle measurements and capacitance value of the primary insulation shall be tested between the primary winding and the capacitance test tap of the insulation of the assembled current transformer.
- (8) Each current transformer shall be tested for composite error as per IEC 60044 based on measuring or protective circuit usage.

## **5.2 Routine Tests**

Routine tests on each transformer shall be as outlined below and in that sequence :

- (1) Verification of terminal markings as per IEC 60044.
- (2) Power frequency tests on secondary windings as per IEC 60044.
- (3) Over voltage interturn tests as per IEC 60044.
- (4) Determination of composite error as per IEC 60044.
- (5) Power frequency dry tests on primary windings as per IEC 60044.
- (6) Partial discharge test as per Clause 5.1 of the specification.
- (7) Dielectric angle measurements as per Clause 5.1 of this specification.

## **6. QUALITY ASSURANCE/QUALITY CONTROL**

The quality assurance/quality control for current transformer shall conform to the requirements included in Part 4, Section 2, Subsection 2.7. The procedures specified shall include, but not be limited to, the followings:

- (1) All routine tests specified type tests performed for this Specification.
- (2) All specified type tests performed for this Specification.

The quality assurance/quality control documentation as specified shall include, but not be limited to, the followings:

- (1) Shop Tests Reports.

(Last Page of E-421)